Machine Learning Analysis of ME/CFS Phenotype Classification

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1 Introduction

Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS) is a chronic, multi-system illness characterized by severe fatigue, brain fog, pain, unrefreshing sleep, and post-exertional malaise (PEM). PEM is a worsening of symptoms after even mild physical or mental activity, often followed by a very slow recovery.

Because ME/CFS symptoms overlap with other conditions and vary strongly across patients, diagnosis is difficult. The goal of this project is to use machine learning to distinguish ME/CFS participants (ME) from healthy controls (CTL) based on survey data and symptom scores, while also identifying which features best capture the underlying ME/CFS pathology.

The report walks through data cleaning and imputation, feature engineering, exploratory data analysis, model selection and hyperparameter search, and finally the evaluation of the best model. Special emphasis is placed on minimizing false negatives (ME patients predicted as CTL), which are the most harmful errors in this medical context.

2 Data Cleaning and Imputation

A large part of the work in this project was careful preprocessing of the dataset to remove leakage, handle missing values, and prepare features for modeling.

2.1 Column Filtering

We first removed variables that were either completely missing or not useful for prediction:

- **Sample_Source** was empty for all rows and was dropped.
- **Study site / source-like columns** were removed later, as they mainly captured where the data were collected rather than the patient's state.

- ME/CFS diagnosis-related variables that could leak the label were dropped, including: *Baseline Sx, Pretest Sx, Starting SSS, Adjusted Mean SSS, Normalized Mean SSS, One Dose Kinetics, MECFS duration ¡ 4, MECFS duration grouped.*

These removals ensure that the model learns from symptoms and recovery patterns rather than from variables that directly encode the diagnosis.

2.2 Handling ME/CFS Duration

The `MECFS duration` column is structurally missing for CTL participants. To avoid injecting spurious information:

- For CTL rows, `MECFS duration` was set to 0.
- For ME rows with missing duration, remaining `NaN` values were also filled with 0.

This makes duration a meaningful feature only for ME participants and safely neutral for controls.

2.3 Symptom Imputation by Participant

Many symptom scores are repeated across multiple visits for the same participant. Before using any model-based imputer, we reused information already observed for that person:

- A symptom list was defined: *Fatigue, Brain Fog, Sore Throat, Tender Nodes, Myalgia, Arthralgia, Headache, Sleep Disturbance, PEM.*
- For each symptom and each `ParticipantID`, missing values were filled using the **within-participant median**. If a participant had at least one non-missing value for a symptom, that median was used to fill any other missing entries for the same person.

This step respects the longitudinal structure of the data: if the same person reported a value on another visit, we prefer that over a generic population-level imputation.

2.4 KNN Imputation for Remaining Numeric Data

After participant-wise filling, some numeric variables still contained missing values. For these we used a KNN-based imputation:

- All numeric columns were collected into a set $\mathcal{N}$.
- A **KNNImputer** with $k = 5$ neighbors was applied so that each missing value is replaced by the mean of its 5 nearest neighbors in feature space.

This allows the imputation to use patterns across multiple features, not only single-column statistics. After this step, the dataset had **no missing values remaining**.

2.5 Symptom Correlation After Imputation

After completing both the participant-level filling and the KNN-based numeric imputation, we computed a correlation matrix focusing specifically on the core symptom variables: Fatigue, Brain Fog, PEM, Sleep Disturbance, and Myalgia. This shows how symptoms relate once the data are fully cleaned.

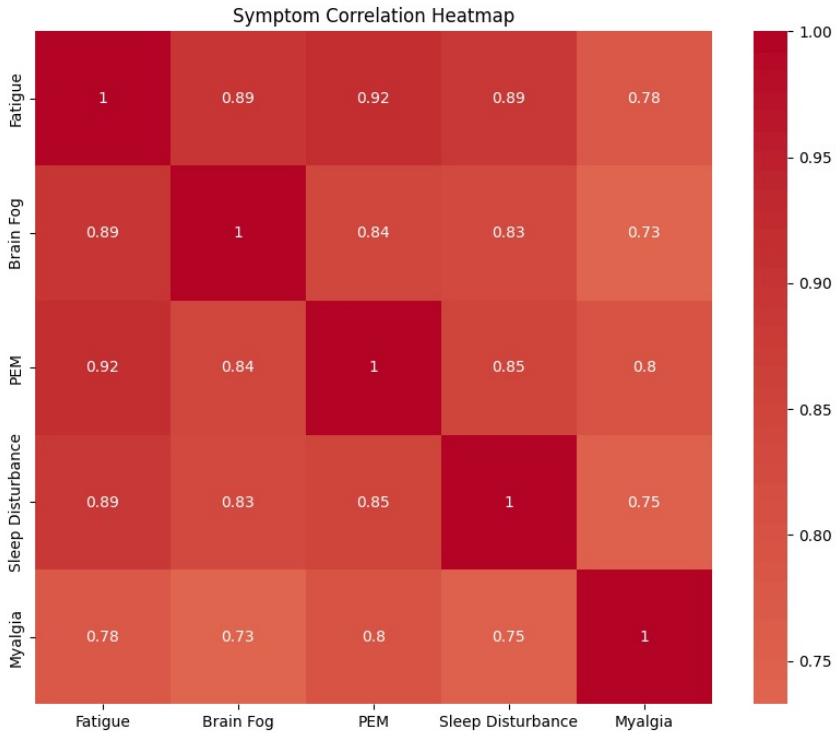

Figure 1: Symptom Correlation Heatmap after Preprocessing

The correlations are very high across all symptom pairs (roughly 0.73 to 0.92). Several points stand out:

- **PEM, Fatigue, and Sleep Disturbance** form the strongest cluster (correlations around 0.85–0.92), capturing the classic "exertion + sleep + fatigue" core of ME/CFS.
- **Brain Fog** is strongly correlated with both physical and neurocognitive symptoms (0.83–0.89), highlighting how cognitive issues co-occur with physical exhaustion.
- **Myalgia** is slightly less correlated, but still shows high values (0.73–0.80) with the other symptoms.

This strong symptom clustering explains why aggregated and interaction features (such as Total Symptom Score and PEM × Recovery) turn out to be so powerful later: many symptoms move together and summarizing them captures most of the relevant information.

2.6 Encoding Categorical Variables

Categorical columns such as `GENDER`, `Day Sx`, `Baseline Sx`, and `Pretest Sx` were initially one-hot encoded using `pd.get_dummies` with `drop_first=True`. As part of the leakage control, diagnosis-related summary variables like `Baseline Sx` and `Pretest Sx` were later removed from the modeling feature set, leaving the model to rely mainly on demographic and raw symptom features.

2.7 Grouping by Participant

Many participants contributed more than one survey record. To avoid data leakage where the same person appears in both training and test sets, we used **ParticipantID as a grouping variable**. This motivated our later choice of GroupShuffleSplit for train/test splitting.

3 Feature Engineering

After cleaning, several engineered features were added to better capture ME/CFS characteristics:

- **Total_Symptom_Score**: sum of key symptoms (Fatigue, Brain Fog, PEM, Sleep Disturbance, and pain variables). This is an overall severity index.
- **Physical_Cluster**: mean of pain-related symptoms (Myalgia, Arthralgia, Tender Nodes).
- **Neuro_Cluster**: mean of fatigue, brain fog, and sleep disturbance, representing a neurocognitive cluster.
- **PEM_x_Recovery** (PEM_Recovery_Interaction): product of PEM score and Recovery Time, capturing how strong the crash is *and* how long it takes to recover.

These features embody clinical intuition: ME/CFS is not just "high PEM" or "long recovery", but the combination of both. This is confirmed later in the permutation importance analysis, where the PEM $\times$ Recovery interaction becomes the single most important predictor.

4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) focused on a few key questions: class balance, demographic differences, and symptom separation between ME and CTL.

4.1 Class Balance and Demographics

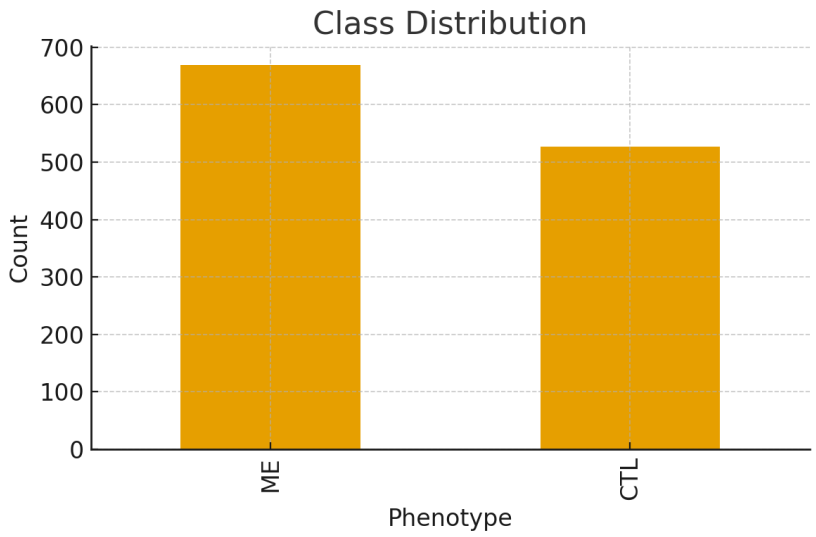

Figure 2: Class Distribution

The dataset is reasonably balanced, with slightly more ME samples than CTL. This helps training and evaluation, as the model does not need to compensate for severe class imbalance.

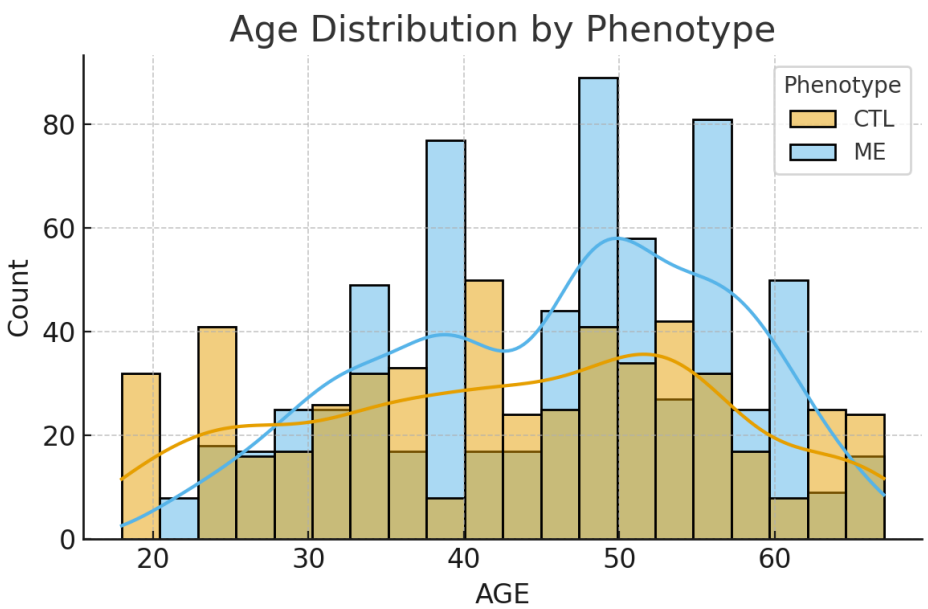

Figure 3: Age Distribution by Phenotype

Age distributions for ME and CTL overlap heavily, with ME participants slightly more concentrated between 40 and 60 years. This suggests that **age alone is not a strong discriminator**.

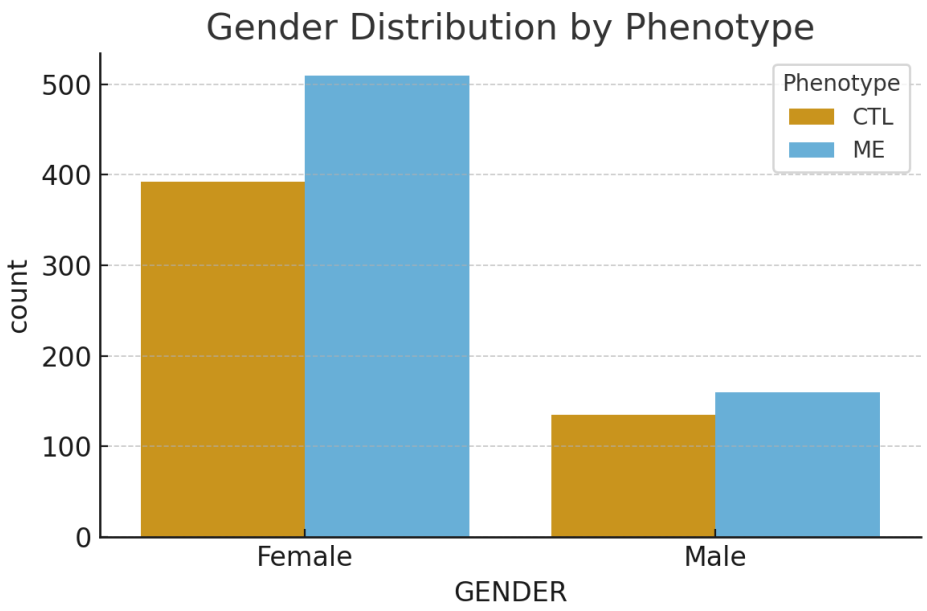

Figure 4: Gender Distribution by Phenotype

Females are more common in both groups, with a higher proportion in the ME group, consistent with ME/CFS epidemiology.

4.2 Symptom Severity by Phenotype

The following plots tell a consistent story: across almost every symptom domain, ME participants report dramatically higher severity than CTL.

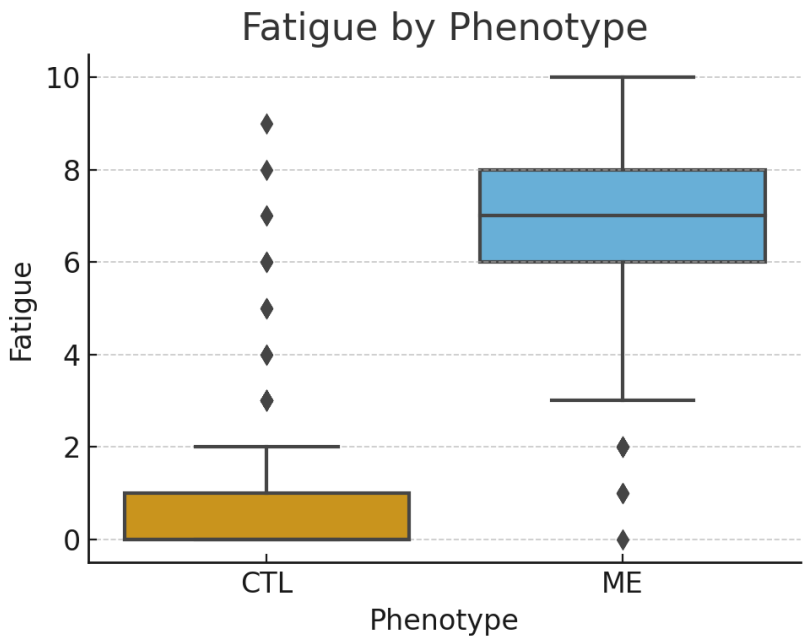

Figure 5: Fatigue by Phenotype

Fatigue levels for CTL are near zero, while ME participants cluster around high values (approximately 7–8). This confirms that pathological fatigue is a core feature of ME/CFS.

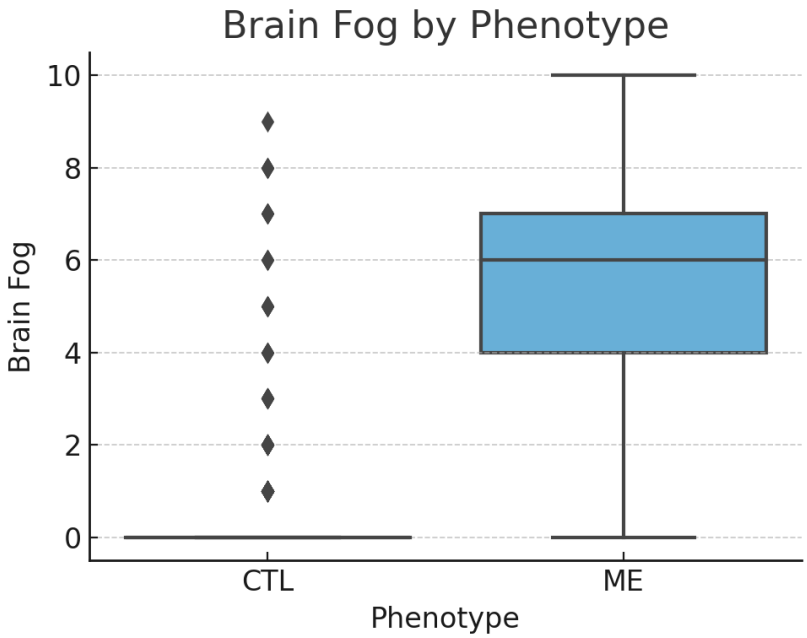

Figure 6: Brain Fog by Phenotype

Brain fog shows a similar separation, capturing cognitive difficulties. CTL values sit near 0, with ME scores shifted strongly upward.

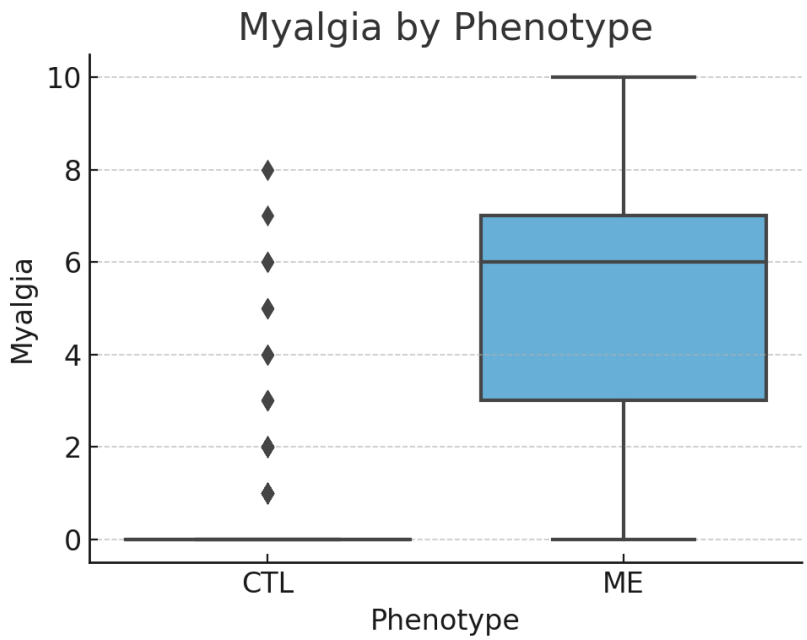

Figure 7: Myalgia by Phenotype

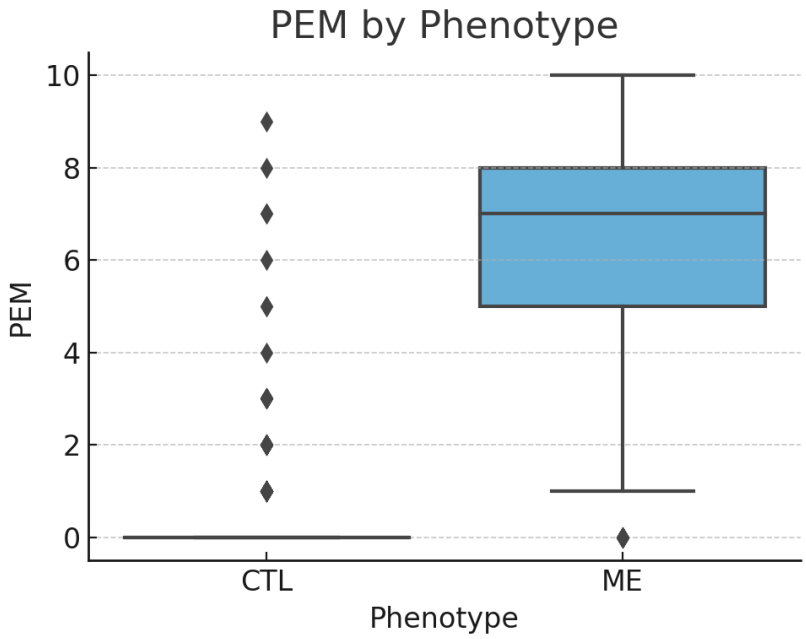

Figure 8: PEM by Phenotype

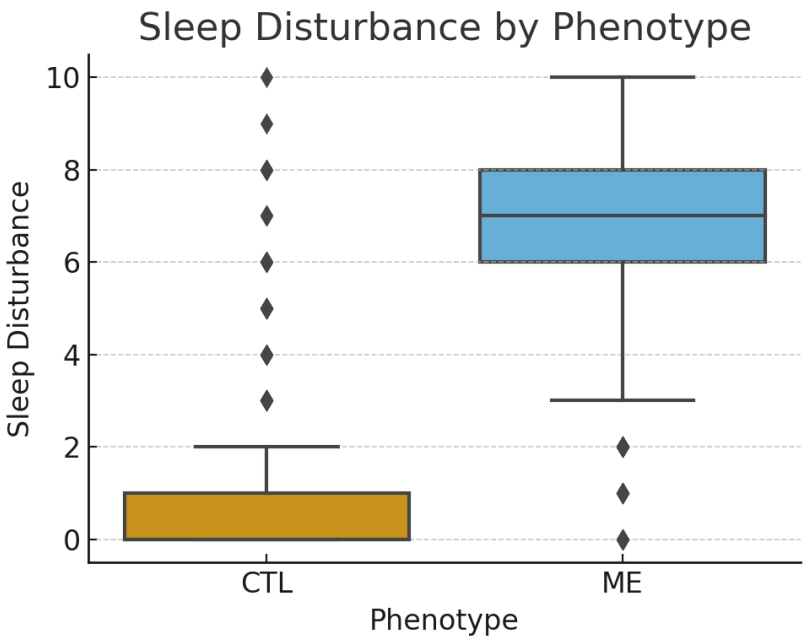

Figure 9: Sleep Disturbance by Phenotype

Myalgia (muscle pain), PEM, and sleep disturbance all show the same pattern: CTL scores near 0, ME scores high and widely spread. Taken together, these plots show that the phenotype label is not random: ME participants form a clear high-symptom cluster across multiple domains.

4.3 Repeated Records Per Participant

In the notebook, we also analyzed how many times each participant repeated the trial. ME participants tended to have more repeated records, with some individuals appearing up to 14 times. This raised a potential risk of data leakage if the same person appeared in both training and test sets, and directly motivated our use of **GroupShuffleSplit** with `ParticipantID` as a grouping variable.

5 Modeling Methodology

5.1 Train/Test Splitting with GroupShuffleSplit

To respect the repeated-measures structure, we used:

- Features: X = all columns except `Phenotype`, `ParticipantID`, and `MECFS duration`.
- Target: y = `Phenotype` (ME vs CTL), label-encoded.
- Groups: groups = `ParticipantID`.

A **GroupShuffleSplit** with test size 20% and random seed 42 was used to create training and test indices such that the same participant never appears in both sets.

5.2 Models and Hyperparameter Search

We tested six model families:

- Random Forest (RF)
- XGBoost (XGB)
- Support Vector Machine (SVM)
- Gradient Boosting Classifier (GBC)
- k-Nearest Neighbors (KNN)
- Logistic Regression (LR)

For each model, a **GridSearchCV** with 5-fold cross-validation on the training set was run, using accuracy as the scoring metric. The parameter grids were:

- **RandomForestClassifier** $n_estimators \in \{100, 200, 300\}$ $max_depth \in \{5, 10, 15, \text{None}\}$ $min_samples_split \in \{2, 5, 10\}$

- **XGBClassifier** $n_estimators \in \{100, 200, 300\}$ $max_depth \in \{3, 6, 9\}$ $learning_rate \in \{0.01, 0.1, 0.2\}$
- **SVC** $C \in \{0.1, 1, 10\}$ $kernel \in \{\text{linear}, \text{rbf}\}$ $\gamma \in \{\text{scale}, \text{auto}\}$
- **GradientBoostingClassifier** $n_estimators \in \{100, 200\}$ $learning_rate \in \{0.05, 0.1, 0.2\}$ $max_depth \in \{3, 4, 5\}$
- **KNeighborsClassifier** $n_neighbors \in \{3, 5, 7, 10, 15\}$ $weights \in \{\text{uniform}, \text{distance}\}$
- **LogisticRegression** $C \in \{0.1, 1, 10\}$ $penalty \in \{\ell_1, \ell_2\}$, solver = `liblinear`

5.3 Two Feature Scenarios

We evaluated two main scenarios:

Case 1: Original Features Engineered columns were dropped, and models were trained on the "true" symptom and demographic features.

Case 2: Engineered Features Only A dataset containing only `Total_Symptom_Score`, `Physical_Cluster`, `Neuro_Cluster`, and `PEM_x_Recovery` was used.

This allowed us to test whether the engineered features could carry most of the predictive power by themselves, and to compare their performance to the original variables.

6 Results

6.1 Cross-Validation Accuracies

CASE 1: TRUE ORIGINAL FEATURES

```
RF: 0.983
XGB: 0.986
SVM: 0.983
GBC: 0.987
KNN: 0.954
LR: 0.976
```

CASE 2: ENGINEERED FEATURES ONLY

```
RF: 0.929
XGB: 0.936
SVM: 0.947
```

```
GBC: 0.927
KNN: 0.944
LR: 0.944
```

Engineered features alone perform surprisingly well, but the best results come from using the full original feature set, with GBC slightly outperforming the other models. We therefore selected a **GradientBoostingClassifier** as the final model.

6.2 Final Test Performance

On the held-out test set (with GroupShuffleSplit by participant), the final model achieved:

```
FINAL TEST RESULTS:
Accuracy: 0.968
ROC AUC: 1.000

Classification Report:
              precision    recall  f1-score   support

         CTL       1.00      0.93      0.96       114
          ME       0.94      1.00      0.97       133

    accuracy                           0.97       247
   macro avg       0.97      0.96      0.97       247
weighted avg       0.97      0.97      0.97       247

Confusion Matrix:
[[106   8]
 [  0 133]]
```

Top 10 feature importances:

```
            feature  importance
10              PEM    0.492918
2           Fatigue    0.303742
15  Baseline Sx_Mid    0.114812
3         Brain Fog    0.032826
11    Recovery Time    0.024408
16   Pretest Sx_Int    0.017466
1        Survey Day    0.004095
```

```
17   Pretest Sx_Low    0.003673
0               AGE    0.003655
8          Headache    0.000766
```

Symptom severity clearly dominates the feature ranking, which is consistent with clinical intuition.

6.3 F-Beta Scores and False Negatives

Because missing an ME patient (false negative) is worse than flagging a healthy control (false positive), we used F-beta scores with $\beta > 1$ to emphasize recall:

```
F-BETA SCORES (Emphasizing False Negatives):

F2-score (beta=2):   0.988
F1.5-score (beta=1.5): 0.982

False Negatives (ME misclassified as CTL): 0
False Positives (CTL misclassified as ME): 8
Sensitivity/Recall: 1.000
Specificity: 0.930
```

Key point: increasing β increases the F-beta score, because our model already has recall = 1.00. When recall is perfect, metrics that give more weight to recall naturally reward the model more. This confirms that **our model handles false negatives extremely well**: in this test set, it does not miss any ME/CFS cases.

6.4 Permutation Importance

Permutation importance with 95% confidence intervals was computed for both the original and engineered feature sets.

PERMUTATION IMPORTANCE (95% CIs):

TRUE Original Features:

```
Recovery Time:   0.161 ± 0.028
Pretest Sx_Int:  0.128 ± 0.030
Baseline Sx_Mid: 0.043 ± 0.020
PEM:             0.013 ± 0.012
Brain Fog:       0.002 ± 0.004
```

TRUE Engineered Features:

```
PEM_x_Recovery:      0.270 ± 0.044
Total_Symptom_Score: 0.153 ± 0.039
Physical_Cluster:    0.002 ± 0.005
Neuro_Cluster:       0.001 ± 0.006
```

Interpretation:

- **Engineered features work better:** PEM_x_Recovery (0.270) is more important than any single original feature.
- **Total_Symptom_Score (0.153)** is also highly informative.
- Among original variables, **Recovery Time (0.161)** and **Pretest Sx_Int (0.128)** contribute the most.
- Individual symptoms (PEM, Brain Fog) have low importance on their own, but become powerful when combined or aggregated.

Overall conclusion from this section: the interaction **PEM × Recovery** is the single most important feature, confirming that the combination of post-exertional crashes and slow recovery captures the ME/CFS phenotype better than any individual symptom.

7 Discussion and Conclusion

This project shows that, after careful cleaning, imputation, and feature engineering, machine learning can robustly distinguish ME/CFS patients from healthy controls.

Key takeaways:

- Symptom severity distributions and engineered interaction features clearly separate ME from CTL.
- A Gradient Boosting model with tuned hyperparameters achieves **96.8% accuracy** and **ROC AUC = 1.0** on a participant-wise held-out test set.
- The model achieves **perfect recall for ME** (no false negatives), which is crucial in a diagnostic support setting.
- F-beta scores increase with β, reflecting the model's strong recall and low rate of missed ME cases.
- Permutation importance highlights **PEM × Recovery** and overall symptom burden as the most informative predictors, aligning well with clinical understanding of ME/CFS.

Future work could extend this approach by including longitudinal patterns, wearable sensor data, or unsupervised clustering to identify ME/CFS subtypes. Nonetheless, the current model already demonstrates that symptom and recovery data, when properly cleaned and engineered, can be used to build a highly reliable classifier that strongly prioritizes avoiding false negatives.